

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title: Smart Complaint & Issue Tracking System

Prepared by

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1. Introduction

1.1 Type of Project

The Smart Complaint & Issue Tracking System is a web-based management system designed to handle internal complaints within an organization.

1.2 Purpose of the Project

The purpose of this system is to:

- Provide a structured platform for submitting complaints.

- Ensure complaints are assigned to appropriate departments.
- Track complaint progress from submission to resolution.
- Maintain transparency through action history logging.
- Generate analytics and reports for administrative monitoring.

1.3 Target Users

The system is designed for:

- **Admin:** Manages users, assigns complaints, and monitors system performance.
 - **Employee:** Submits complaints and tracks their status.
 - **Staff:** Resolves complaints within assigned departments.
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2. Functional Requirements

The system shall provide the following functionalities:

2.1 User Management

1. The system shall support role-based access (Admin, Employee, Staff).
2. The Admin shall create user accounts.
3. The Admin shall update user accounts.
4. The Admin shall deactivate user accounts.
5. Users shall log in and log out securely.

2.2 Complaint Management

6. Employees shall submit complaints with a title, description, category, and priority.
7. Employees shall edit complaints before assignment.
8. Employees shall view their submitted complaints.
9. Admin shall assign complaints to departments (HR, IT, Finance, Marketing & Sales, Software & Product Development).
10. Staff shall view department-specific complaints.
11. Staff shall update the complaint status (Pending, In Progress, Resolved).
12. Staff shall add resolution notes.

2.3 Tracking & Logging

13. The system shall maintain a complaint action history.
14. The system shall calculate the resolution time.
15. The system shall support priority escalation.

2.4 Attachments

16. Users shall upload complaint attachments.
17. Users shall delete uploaded attachments.

2.5 Notifications

18. The system shall send automated notifications when:
 - The complaint is assigned
 - Status is updated

- The complaint is resolved

2.6 Reports & Analytics

19. Admin shall view the dashboard summary.

20. System shall generate:

- Department-wise reports
- Priority-wise reports
- Average resolution time report

2.7 Search & Filter

21. The system shall allow searching for complaints by:

- Keyword
 - Category
 - Status
 - Priority
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3. Non-Functional Requirements

- Passwords must be encrypted.
- Role-based access must restrict unauthorized actions.
- Only Admin can manage users.

- The system should respond within 2–3 seconds.
 - Reports should be generated within 5 seconds.
 - The interface should be simple and user-friendly.
 - Dashboard should display clear statistics.
 - Complaint data must not be lost.
 - Action history must remain consistent.
 - The system should handle an increasing number of users and complaints.
 - Strictly follows MVC Architecture
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4. Class Diagram

 CSE470_Project_Group_8

5. Tools and Technologies

5.1 Frontend

The frontend of the system is developed using:

- **JavaScript** – Core programming language for client-side logic.
 - **React.js** – A JavaScript library used to build interactive and dynamic user interfaces.
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5.2 Backend

The backend of the system is developed using:

- **Node.js** – JavaScript runtime environment used to execute server-side code.
 - **Express.js** – A lightweight web framework for Node.js.
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5.3 Database

- **MongoDB** – A NoSQL document-based database.

5.4 Other Tools

- VS Code – Code editor
 - draw.io – Class Diagram
 - Git & GitHub – Version control
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6. Compatibility with System Environment / OS

The system is web-based and compatible with:

- Windows 10 / 11
- Linux
- macOS

Supported Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Since it runs on a browser, it does not depend heavily on OS, only requires:

- Web browser
 - Internet or local server environment
-

7. Implementation

To be updated with each Sprint.

20 Features:

- Role-based access control (Admin, Staff, Employee)
- Admin-managed user account creation

- User account update and deactivation
 - Complaint submission with title, description, category, and priority
 - Attachment upload for complaint evidence
 - Complaint status tracking (Pending, In Progress, Resolved)
 - Complaint editing before assignment
 - View personal complaint history
 - Assign complaints to departments (IT, HR, Facilities, etc.)
 - Department-specific complaint inbox
 - Update complaint status during resolution
 - Add resolution comments and notes
 - Maintain a complaint action history log
 - Admin dashboard showing overall complaint summary
 - Department-wise complaint report
 - Priority-wise complaint statistics
 - Average resolution time analytics
 - Complaint search and filter functionality
 - Automated notifications
 - Priority escalation system
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8. Challenges

- Designing proper role-based access control.
 - Maintaining a complaint history without redundancy.
 - Implementing automated notifications.
 - Designing clean UML relationships.
 - Maintaining MVC Structure.
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9. Conclusion

The Smart Complaint & Issue Tracking System successfully provides a structured and transparent method to manage internal organizational complaints. It improves accountability, efficiency, and tracking of issues while generating meaningful analytics for management.

The system fulfills all 20 required features and demonstrates practical implementation of object-oriented design and database management concepts.

10. References

1. Software Engineering Lecture Slides (CSE470)
2. UML Documentation – draw.io
3. W3Schools

SPRINT 1

(User Creation + complaint submission)

1. Admin-managed user account creation
2. Role-based access control
3. User account update and deactivation
4. Complaint submission
5. Complaint status tracking

Feature 1: Admin-managed user account creation

- user created role wise (Staff & Employee)
- user (admin) created by developer

The image shows a 'User Management' interface. At the top, there are tabs for 'Employees' and 'Staff', and a '+ Create User' button. Below this is a form with fields for 'Name', 'Email', 'Password', and a role dropdown menu set to 'Employee', followed by a 'Submit' button. Below the form is a table with columns 'Name', 'Email', 'Status', and 'Action'. The table contains two rows: 'Test Staff' with email 'staff1@example.com' and status 'active', and 'Modern staff' with email 'staff2@example.com' and status 'inactive'. The 'Action' column for 'Test Staff' has 'Edit' and 'Deactivate' buttons, while for 'Modern staff' it has an 'Edit' button.

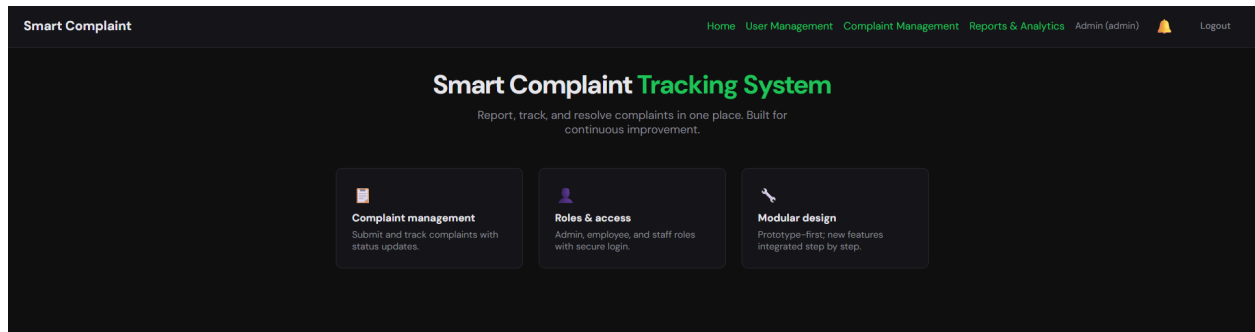
Name	Email	Status	Action
Test Staff	staff1@example.com	active	Edit Deactivate
Modern staff	staff2@example.com	inactive	Edit

Feature 2: Role-based access control

Admin:

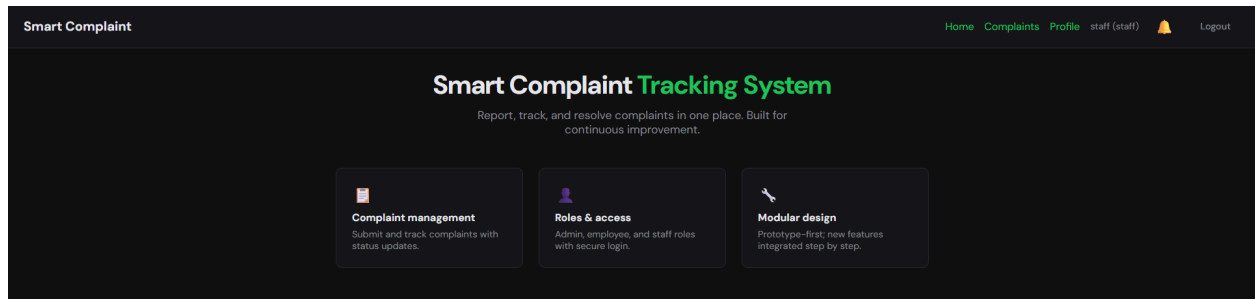
- can manage and see users

- manage complaints
- see report and analytics



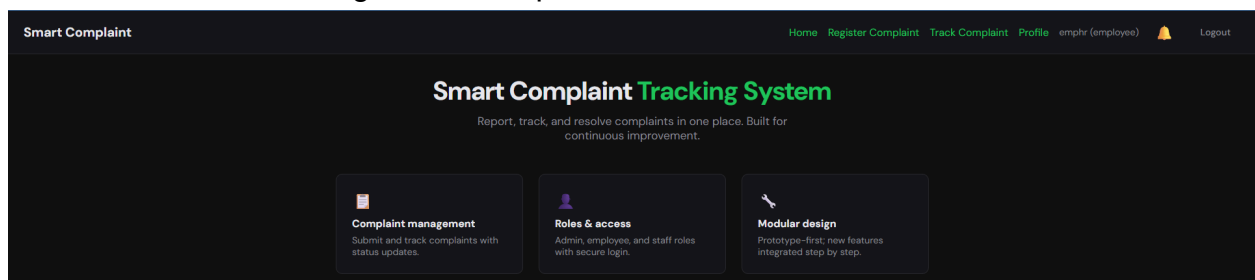
Staff:

- can see complaints assigned to them



Employee:

- can register complaint
- track his registered complaints



Feature 3: User account update and deactivation

- change user name, email, department
- deactivate user (user status changes to inactive with red mark)

User Management

All Admin Employees Staff + Create User

Edit User

Admin

▼

Select Department

▼

Update

Name	Email	Department	Status	Action
Admin	admin@example.com	-	active	Edit Deactivate
emphr	emp@hr.com	HR	active	Edit Deactivate
empit	emp@it.com	IT	active	Edit Deactivate
staff	staff@example.com	-	active	Edit Deactivate
empfin	emp@fin.com	Finance	active	Edit Deactivate

Feature 4: Complaint submission

- only employee can submit complaints

Submit Complaint

Select Category

▼

Attachments (optional)
Images (JPG, PNG, GIF, WEBP), PDF, DOC, DOCX – Max 10MB each, up to 5 files

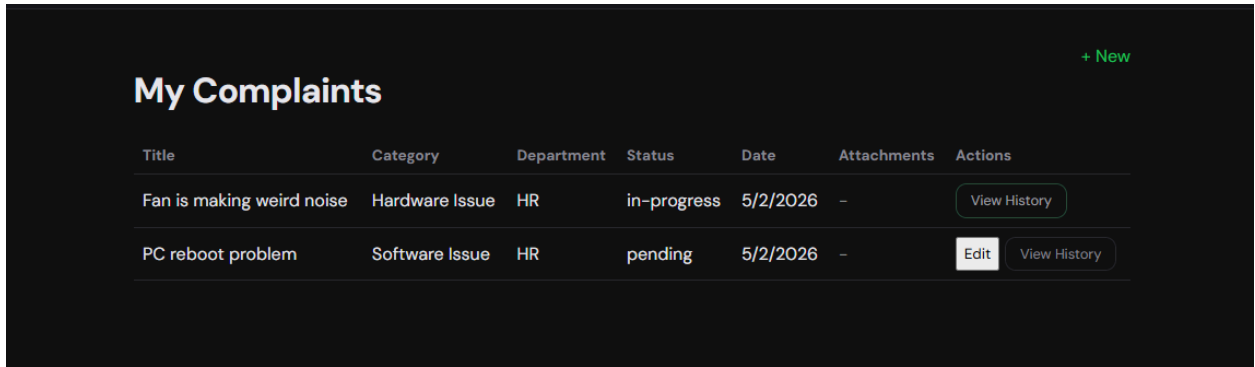
Choose Files

No file chosen

Submit

Feature 5: Complaint status tracking

- employee can track his own complaints



Title	Category	Department	Status	Date	Attachments	Actions
Fan is making weird noise	Hardware Issue	HR	in-progress	5/2/2026	-	View History
PC reboot problem	Software Issue	HR	pending	5/2/2026	-	Edit View History

Sprint 2 (Complaint Management)

- 1.Complaint editing before assignment
- 2.View personal complaint history
- 3.Assign complaints to departments
- 4.Department-specific complaint inbox
- 5.Update complaint status during resolution

1.Complaint editing before assignment

Here, we can update the complaint after submitting and before assignment.

My Complaints

+ New

Title	Category	Status	Date	Actions
sample complaint	Marketing & Sales	pending	4/16/2026	this is a complain Save Cancel

Now editing the complaint:

My Complaints

+ New

Title	Category	Department	Status	Date	Actions
this is a complain	Hardware Issue	Not assigned	pending	4/16/2026	sample complain edited Save Cancel

before:

this is a complain

sample complain

Category: Hardware Issue

Status: pending

Update Status:

Pending

Created by: Employee One

Created on: 4/16/2026, 8:20:40 PM

Select Staff

Assign to Staff

After:

this is a complain

sample complain edited

Category: Hardware Issue
Status: pending

Update Status: Pending

Created by: Employee One
Created on: 4/16/2026, 8:20:40 PM

Select Staff

Assign to Staff

2.View personal complaint history

Employees can view their personal complaint history here.

+ New

My Complaints

Title	Category	Department	Status	Date	Actions
this is a complain	Hardware Issue	Not assigned	pending	4/16/2026	Edit

3. Assign complaints to departments

Here , we can assign complaints to individual departments.

employee 2 com

employee 2 submitted a com to check if another em can see

Category: HR
Status: pending

Update Status: Pending ▼

Created by: Employee Two
Created on: 4/16/2026, 6:26:47 PM

Select Staff ▼

Select Staff

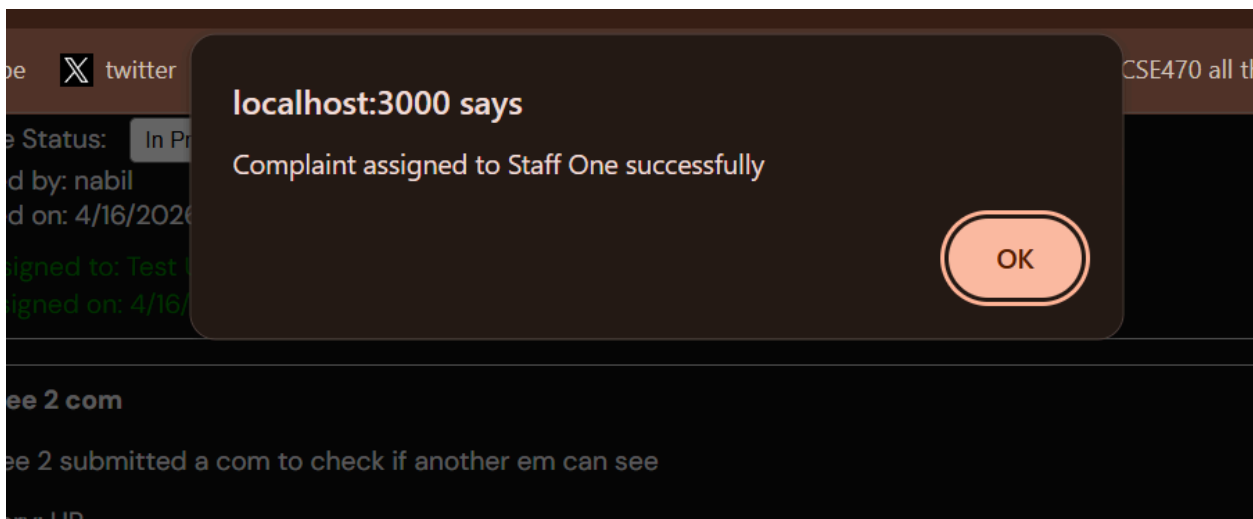
Test User (testuser@test.com)

fahim (fahim@yahoo.com)

Staff One (staff1@example.com)

Staff Two (staff2@example.com)

Assign to Staff



then , after logging into staff1@example.com , we see:

Smart Complaint Home Complaints Profile Staff One (staff) Logout

Complaints

My Assigned Complaints

employee 2 com

employee 2 submitted a com to check if another em can see

Category: HR
Current Status: pending
Created by: Employee Two
Created on: 4/16/2026, 6:26:47 PM

Update Status: Pending ▼

4.Department-specific complaint inbox

This is a detailed view of complaint lists.

Smart Complaint

[Home](#) [User Management](#) [Complaint Management](#) [Reports & Analytics](#) [Admin \(admin\)](#) [Logout](#)

Complaints

Unassigned Department (5)

hello

this is a complaint edited

Category: HR

Status: pending

Update Status: Pending

Created by: I

Created on: 3/27/2026, 4:57:24 PM

Select StaffAssign to Staff

hh

knagjadjnndgbv

Category: IT

Status: pending

Update Status: Pending

Created by: I

Created on: 3/27/2026, 4:58:49 PM

Select StaffAssign to Staff

sample complaint

this is a complain

5.Update complaint status during resolution

Here , after submitting complaints , the admin can update the status of complaints.

First , we put status to **ASSIGNED**

this is a complain

sample complain edited

Category: Hardware Issue

Status: assigned

Update Status: Assigned

Created by: Employee One

Created on: 4/16/2026, 8:20:40 PM

✔ Assigned to: Staff One

📅 Assigned on: 4/16/2026, 8:27:54 PM

Then , on the *employee1* track complaint option , we see status as **ASSIGNED** .

My Complaints + New					
Title	Category	Department	Status	Date	Actions
this is a complain	Hardware Issue	Not assigned	assigned	4/16/2026	

Now after changing to **In Progress**:

this is a complain

sample complain edited

Category: Hardware Issue
Status: in-progress

Update Status:

Created by: Employee One
Created on: 4/16/2026, 8:20:40 PM

☒ Assigned to: Staff One
☐ Assigned on: 4/16/2026, 8:27:54 PM

We see:

My Complaints + New					
Title	Category	Department	Status	Date	Actions
this is a complain	Hardware Issue	Not assigned	in-progress	4/16/2026	

Sprint 3 (Workflow + Enhancements)

1. Add resolution comments and notes
2. Maintain a complaint action history log
3. Attachment upload
4. Complaint search and filter
5. Automated notifications

SS-01 (Feature 1+2 = COMMENT + HISTORY)

Task-1

Description:

Category: N/A

Priority: medium

Status: pending

Created by: Employee One (employee1@example.com)

Created on: 5/2/2026, 2:44:27 PM

Update Status

Pending ▼

Optional comment with status change

Add Comment

Write your comment here...



Post Comment

Activity Timeline

Activity Timeline

2 actions



Admin User ADMIN

Just now

Admin User added a comment



"Looking into this"



Employee One EMPLOYEE

4 minutes ago

Employee One created this complaint



Complaint submitted by Employee One

 Network Problem

 medium

SS-02 (Feature 3 Attachment upload)

Submit Complaint

Task-2

Attach Complaint



Policy Suggestion 

Attachments (optional)

Images (JPG, PNG, GIF, WEBP), PDF, DOC, DOCX – Max 10MB each, up to 5 files

Choose Files

project work follows.pdf

project work follows.pdf

Remove

Submit



Complaints

All Categories

pending

All Priority

IT

Search

Reset

Found 2 complaint(s) [Show all](#)

IT Department (2)

Task-2

Attach Complaint

Category: Policy Suggestion

Status: pending

Update Status:

Pending

Created by: Employee One

Created on: 5/2/2026, 2:55:12 PM

Attachments (1):

project work
follows.pdf
66.4 KB

Staff One (staff1@example.com)

Assign to Staff

View History

Sprint 4 (Priority and Analysis)

1. Priority Escalation System - Increases Priority but cannot be increased after Priority = critical.
2. Admin Dashboard - Shows the Report and Statistics.
3. Department-wise Report - Shows report and chart for each department.
4. Priority-wise Statistics - Shows statistics for low, medium, high and critical priorities.
5. Average Resolution time Analytics - Calculates average resolution time and rate for each department and priorities.

1. Priority Escalation System

Before and after Escalation

Printer C is not working!

Ink is finished.

Category: Hardware Issue
Status: pending

Update Status: Pending ▼

Priority: **low**

Escalate

Created by: Employee One
Created on: 5/9/2026, 10:56:21 AM

Select Staff ▼ Assign to Staff

View History

Printer C is not working!
 Ink is finished.
 Category: Hardware Issue
 Status: pending
 Update Status: Pending ▼
 Priority: **high**
Escalate
 Created by: Employee One
 Created on: 5/9/2026, 10:56:21 AM

Select Staff ▼
Assign to Staff

View History

No more escalations after critical

Printer C is not working!
 Ink is finished.
 Category: Hardware Issue
 Status: pending
 Update Status: Pending ▼
 Priority: **critical**
 Created by: Employee One
 Created on: 5/9/2026, 10:56:21 AM

Select Staff ▼
Assign to Staff

View History

2. Admin Dashboard

Admin Dashboard

Department-wise Report

Department	Total	Pending	Assigned	In Progress	Resolved	Avg Resolution (Hours)
HR	0	0	0	0	0	N/A
IT	3	3	0	0	0	N/A
Finance	0	0	0	0	0	N/A
Marketing & Sales	0	0	0	0	0	N/A
Software & Product Development	7	1	0	4	2	271.3

Priority-wise Statistics

Priority	Count	Percentage
low	1	10.0%
medium	2	20.0%
high	2	20.0%
critical	5	50.0%
Total	10	100%

Average Resolution Time Analytics

This analytics section shows average resolution times for complaints overall, by department, and by priority.

271.3 hours

Overall Avg Resolution

2

Resolved Complaints

20%

Resolution Rate

3. Department-Wise Report

Department-wise Report

Department	Total	Pending	Assigned	In Progress	Resolved	Avg Resolution (Hours)
HR	0	0	0	0	0	N/A
IT	3	3	0	0	0	N/A
Finance	0	0	0	0	0	N/A
Marketing & Sales	0	0	0	0	0	N/A
Software & Product Development	7	1	0	4	2	271.3

Visual Chart

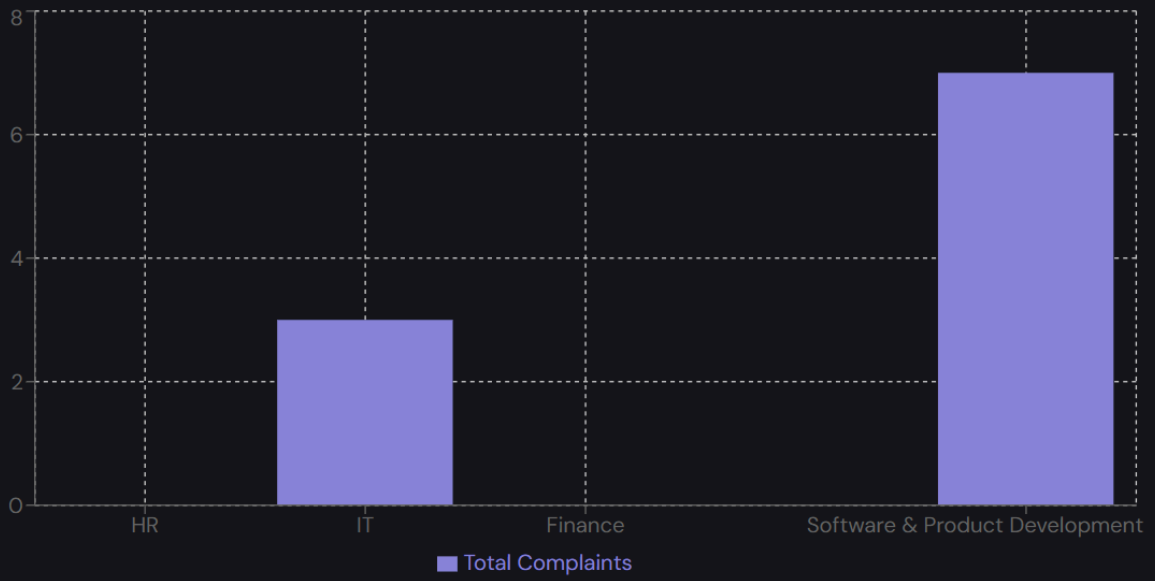
Total Complaints

Pending

Assigned

In Progress

Resolved



Visual Chart

Total Complaints

Pending

Assigned

In Progress

Resolved



4. Priority-Wise Statistics

Priority-wise Statistics		
Priority	Count	Percentage
low	1	10.0%
medium	2	20.0%
high	2	20.0%
critical	5	50.0%
Total	10	100%

5. Average Resolution Time Analytics

Average Resolution Time Analytics		
This analytics section shows average resolution times for complaints overall, by department, and by priority.		
271.3 hours Overall Avg Resolution	2 Resolved Complaints	20% Resolution Rate
Average Resolution by Department		
Department	Avg Resolution Time (Hours)	Resolved
HR	N/A	0
IT	N/A	0
Finance	N/A	0
Marketing & Sales	N/A	0
Software & Product Development	271.3	2

Average Resolution by Priority		
Priority	Avg Resolution Time (Hours)	Resolved
low	N/A	0
medium	541.8	1
high	N/A	0
critical	0.8	1